# Content.

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Let $k$, $p$ and $n$ be integers such that $0 \leq k \leq k+p < n$. The symmetric shift register $\theta : \{0,1\}^n \to \{0,1\}^n$ with respect to $k$, $p$ and $n$ is defined by $\theta(a_1 \cdots a_n) = a_2 \cdots a_{n+1}$ where

$a_{n+1} = a_1'$ if $k \leq a_2 + \cdots + a_n \leq k+p$, and $a_{n+1} = a_1$ otherwise,

where $a_1' = 0$ if $a_1 = 1$, and $a_1' = 1$ if $a_1 = 0$.

We let $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ be the string generated from $A$ by the symmetric shift register with parameters $k$, $p$ and $n$. For $i \geq 0$, then

$a_{n+i+1} = a_{i+1}'$ if $k \leq a_{i+2} + \cdots + a_{i+n} \leq k+p$. Otherwise, $a_{n+i+1} = a_{i+1}$.

The minimal period of $A^\infty$ is called the minimal period of $A$ with respect to the symmetric shift register with parameters $k$, $p$ and $n$.

The goal is to determine the minimal periods in the general case.

**Menu option 1.** In this alternative the main period is determined by an inductive method based on the article

"Representation of symmetric shift registers" by Jan Søreng, published and available at arXiv [2505.23974].

This method can not be used for all type of cases. However, the method can handle very large periods. Moreover, some intermediate parameters and calculations are displayed. These intermediate results can be studied to get information of how the periods are constructed and the inner structure of the strings generated. ( This alternative is also available at the front page ).

**Menu option 2.** In this alternative the minimal period is determined by a direct approach based on the definition of the given symmetric shift register. This method handle all types of start conditions. However, the method has certain limitations since the direct method is computationally intensive. If the periods are too large, we can use menu option 3.

**Menu option 3.** This alternative handles the general case also for large periods. This will be done by a method based on the inductive method given in menu option 1. In order to use the inductive method some assumptions must be fulfilled. The first step is therefore to reformulate the problem such that these conditions are fulfilled. Then the minimal period can be determined by the inductive method ( menu option 1 ). This method is very fast and can compute very large periods.

**Notation.** If $j \geq 1$, then $0_j$ denotes $j$ consecutive 0's, and $1_j$ denotes $j$ consecutive 1's. Moreover, if $j = 0$, then $0_j$ and $1_j$ denote the empty string. Suppose $Q = (q_1, \cdots, q_s)$ where $s$ is even, $q_1 \geq 0$, $q_i > 0$ for $1 \leq i < s$, and $q_s \geq 0$. Then $Q$ is the even vector representation of the string

$$A = 1_{q_1} 0_{q_2} 1_{q_3} 0_{q_4} \cdots 1_{q_{s-1}} 0_{q_s}.$$

For example $Q = (2, 3, 1, 2)$ represents $A = 1_2 0_3 1_1 0_2 = 11000100$. Moreover, $Q = (4, 2, 3, 0)$ represent $A = 1_4 0_2 1_3 0_0 = 111100111$.

**Remark.** The description of the input is available in the user guide for each menu option. We will mention only one point. The input string $A$ is given as an even vector $Q = (q_1, \cdots, q_s)$ as described above. But, in menu option 1 we must assume that $q_1 > 0$.